Renal, Nephrology & Transplant

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 88	Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter
Ch 90	Fundamentals of Nephrology
Ch 91	Renal Transplantation
★	Putting it together: kidney function and transplant decisions

88 Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter

1 · Kidney: position and relations

- ≈**11 cm**, 125–170 g (male); T12–L3, **right 1–2 cm lower**; descends ≈3 cm on inspiration/standing.
- P1 Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal mass — *relations, coverings, Gerota open below, hilum order*
- Traction tears liver (hepatorenal ligament) or splenic capsule (splenorenal ligament). Posterior also: iliohypogastric, ilioinguinal nerves, pleura.
- P1 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL — *access hazards: pleura, lung, liver/spleen, retrorenal colon*
- Upper-pole excision via diaphragm → pneumothorax.

2 · Coverings and compartments

- Perirenal fat enters the sinus. Each perirenal space separate (endourologic extravasation rarely crosses); **anterior pararenal space** (colon, duodenum, pancreas) crosses the midline; posterior pararenal = fat only.
- Transperitoneal: incise **white line of Toldt**; right hilum — **Kocher manoeuvre**; left — mobilise pancreatic tail, spleen, splenic vessels medially.

3 · Parenchyma and its imaging

- Cortex ≈1 cm. **Columns of Bertin** carry interlobar arteries → track **through pyramid and papilla**.
- GFR ≈**125 mL/min**; ≈99% of filtrate reabsorbed. Urothelium calyces → urethra → urothelial carcinoma at any level.

Modality	Normal kidney
US	Cortex iso-/hypoechoic to liver (hyperechoic in newborn)
CT	Unenhanced 30–60 HU, cortex = medulla; enhanced 80–120 HU
MRI	T1 cortex brighter; T2 cortex slightly darker

CT phase	Timing	Use
Arterial	20–30 s	Vessels
Corticomedullary	30–70 s	Cortex enhances
Nephrographic	**80–120 s**	**Best for renal masses**
Excretory	>3 min	Collecting system, ureter

- IVU: nephrotomogram 60–90 s, panoramic 15 min; no excretion at 24 h = non-functioning kidney.
- **Corticomedullary differentiation** lost in CKD and AKI, returns with recovery; parenchymal volume tracks function better than length.

Congenital anomalies

- **Malrotation**: pelvis central; calyces medial to pelvis = hallmark. **Ectopia**: ureter length fits; poor drainage → stasis, infection, stones.
- **Crossed fused ectopia**: crossed ureter returns to a normal orifice.
- **Horseshoe**: lower poles fused; ascent arrested by the **IMA**; UPJO, stones, infection, multiple arteries.

4 · Renal vessels, lymphatics and nerves

- P2 Renovascular hypertension and ischaemic nephropathy: renal vascular anatomy, pathophysiology, evaluation and treatment — *segmental end arteries, accessory 25–28%, Brödel line, veins; left renal vein **6–10 cm**, nutcracker, circumaortic*
- Brödel line site varies — map by angiography or methylene blue (Boyce anatrophic nephrolithotomy) → *Ch 91*.
- **CT angiography** = gold standard for vascular anatomy (misses vessels <2 mm).
- Lymph: left → left para-aortic; right → interaortocaval/paracaval, **may cross to left para-aortic**.

5 · Percutaneous puncture and vessels

- P1 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL — *Sampaio puncture table: upper infundibulum 67%, fornix no arterial injury*
- Through-and-through puncture → anterior segmental vessels → marked haemorrhage.

> **KEY POINT**
> Why the fornix? Arteries run on the anterior and posterior faces of the infundibula; a forniceal puncture through the papilla avoids them. If an infundibulum is unavoidable, choose the lower pole. Vascular injury → haemorrhage, AV fistula, pseudoaneurysm, loss of function.

6 · Pelvicalyceal system

- P1 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL — *Sampaio types, duplex pelvis, perpendicular and crossed calyces*
- Crossed calyces: interpelviocalyceal region on fluoroscopy = indirect sign.
- Dilatation mimics: extrarenal pelvis; **megacalycosis** (non-obstructive, extra calyces, normal pelvis); calyceal diverticulum (fornix into cortex).

7 · Ureter: course and relations

- Adult 22–30 cm (neonate 6.5–7 cm). **Narrowings** (stone impaction): UPJ, iliac crossing, UVJ.
- On psoas; crosses the **common iliac bifurcation** (landmark). Gonadal vessels cross anteriorly; right: ileum, appendix; left: sigmoid mesocolon. **Pelvic brim** a commonest injury site.
- Female: posterior boundary of ovarian fossa → under the **uterine artery 1.5–2 cm** from cervix ("water under the bridge"); left more at risk in vaginal surgery. Male: vas loops over it.
- **Intramural** 1.2–2.5 cm in Waldeyer sheath; bladder pressure closes oblique tunnel → antireflux.

8 · Radiologic anatomy of the ureter

- **Peristalsis**: whole ureter rarely opacified; continuous column → obstruction or atony.
- Stricture needs **proximal dilatation**; crossing vessels and atherosclerotic iliac indentation mimic it. Prostatic enlargement → "fish-hook" distal ureter.
- Incomplete duplication (one bud splits, one orifice) commoner than complete (two buds, two orifices).

> **KEY POINT**
> **Weigert–Meyer rule**: upper-moiety orifice **inferior and medial** → ureterocele, obstruction; lower moiety → VUR.

- Medial displacement: bilateral — retroperitoneal fibrosis, pelvic lipomatosis, post-APR; unilateral — hypogastric nodes/aneurysm, bladder diverticulum.

9 · Ureteric blood supply, nerves, histology

Segment	Artery	Enters	Endoureterotomy
Proximal	Renal	Medially	**Lateral**
Middle	Common iliac	Posteriorly	**Anterior**
Distal	Superior vesical	Laterally	**Medial**

- Lymph: abdominal ureter → para-aortic/paracaval; pelvic → iliac.
- Fragile longitudinal anastomosis: preserve adventitia, avoid lateral retraction (ischaemic stricture).
- **Myogenic peristalsis** (minor-calyx pacemakers): excised ureter still contracts; denervation does not cause reflux. Pain T11–L1 (S2–S4 lower) → flank, groin, scrotum/labia.
- **UPJO**: excess longitudinal muscle, collagen.

10 · Endoscopic anatomy

- Trigone: base = interureteric ridge (Mercier bar); most vascular part. Orifices 1–2 cm from midline, move laterally on filling.

Lyon axis	Grades
Position	A medial trigone (normal) · B · C lateral
Configuration	0 volcano (normal) · 1 stadium · 2 horseshoe · 3 golf-hole

- Higher grade → more reflux. **Intramural ureter narrowest** (3–4 mm); may need dilatation.
- Brim and UPJ look stenotic but open with irrigation; iliac pulsation marks the brim, where ureter angles posteriorly.
- Lower-pole infundibulum needs ≈**140°** deflection. Ducts of Bellini dilate with obstruction.

> **ALGORITHM**
> Infundibulotomy: superior → inferior → posterior quadrants; **never anterior**

90 Fundamentals of Nephrology

1 · Why a urologist needs nephrology

- **Nephron**: ≈ **1 million** per kidney; number fixed at birth; reduced nephron mass (nephrectomy, agenesis) → **hyperfiltration** of the rest.
- **Erythropoietin** from interstitial fibroblasts (hypoxia); **calcitriol** by proximal 1α-hydroxylation (PTH, low Ca/phosphate).

> **KEY POINT**
> Surgery, obstruction, its relief and drug dosing all change volume, electrolytes and acid–base. Glomerulus = pressure filter set by arteriolar tone; tubule = energy-hungry, near-hypoxic — most disorders fail one or the other.

2 · Measuring kidney function

- P2 GFR estimation: clinical assessment and in the prospective transplant donor — *inulin/iohexol gold standard; creatinine secreted (24-h clearance overestimates); CKD-EPI 2021 race-free; cystatin C; combined equation most accurate*
- Use the combined equation when creatinine eGFR is biased or a decision sits at a staging threshold.
- SCr falsely ↑: muscle, meat, creatine; **trimethoprim, cimetidine**, fenofibrate (block secretion). Cystatin C ↑: hyperthyroidism, steroids, inflammation.

> **KEY POINT**
> Trap: after cystectomy or bladder repair, rising SCr with ascites may be **intraperitoneal urine leak** reabsorbed across the peritoneum — not kidney injury.

- P1 Proteinuria — *normal <* ***150 mg/day****, nephrotic >3.5 g/day; spot early-morning ACR first, PCR for non-albumin protein; dipstick misses light chains*
- Albuminuria predicts CKD progression, CVD, death.
- **eGFR invalid when function is changing** (anuric = GFR zero) → AKI defined by SCr change + urine output; both late; novel biomarkers investigational.

3 · Acute kidney injury

Glomerular haemodynamics

- Net filtration pressure ≈ **10 mm Hg** = glomerular hydrostatic 60 − Bowman's 18 − oncotic 32.

Arteriolar change	RBF	GFR
Afferent constriction	↓	↓
Afferent dilatation	↑	↑
Efferent constriction	↓	↑
Efferent dilatation	↑	↓

Efferent constriction lowers flow but raises GFR — so angiotensin II blockade drops GFR when perfusion is low.

Definition and staging (KDIGO)

Stage	Serum creatinine	Urine output
1	Rise ≥ **0.3 mg/dL** (within 48 h†) or ×1.5 baseline (within 7 days†)	<0.5 mL/kg/h for 6 h
2	×2 baseline	<0.5 mL/kg/h for 12 h
3	×3, or SCr ≥4 mg/dL with acute rise ≥0.5 (†or dialysis started)	<0.3 mL/kg/h for 24 h, or anuria 12 h

- P2 Acute renal failure (AKI) and acute tubular necrosis — *causes, ATN mechanism, AEIOU dialysis indications*
- **3 months** divides AKD from CKD. Intratubular: myeloma casts, crystals.

Prerenal AKI

- Bland urine, low FeNa, **immediate recovery** on correction.
- **Autoregulation**: **myogenic reflex** — prostaglandin afferent dilatation, **disabled by NSAIDs**; **TGF** — macula densa low NaCl → renin → angiotensin II → efferent constriction, **disabled by RAAS blockade**. Impaired by age, CKD, transplant, sepsis, contrast.
- **Modifiable**: hypotension, anaemia, sepsis, nephrotoxins, rhabdomyolysis, hypercalcaemia.
- P2 Acute renal failure (AKI) and acute tubular necrosis — *prerenal vs ATN indices: urine Na 20, FeNa <**1%**, FeUrea <**35%**, SG, muddy-brown casts, fluid response*
- High FeNa after diuretics ≠ ATN — use FeUrea. **Response to correction is the single most important test.**
- MAP >**65 mm Hg**; elderly vasculopaths need more.

Fluids (mOsm/L): plasma and 0.9% saline 308; Plasmalyte 294; Ringer's lactate 273; 0.45% saline 154; D5 0 (free water).

Intrinsic AKI — the cast localises the compartment

	ATN	Acute GN	AIN
Cause	Ischaemia, toxins	Immune complex, anti-GBM, cell-mediated	Drug, infection, systemic
Urine	Granular casts	Dysmorphic RBCs, **red cell casts**, proteinuria	WBCs, **white cell casts** ± eosinophils

Acute tubular necrosis

- **Rhabdomyolysis** (obese, long flexed lateral robotic case; may be painless) → **early aggressive isotonic fluid**; alkalinisation unproven. †Prevent: padding, limit flexion and time.
- **Diuretics only for symptomatic overload; they do not treat ATN.**

Postcontrast AKI

- SCr rise within **48 h** of IV iodinated contrast. Risk: age, diabetes, CKD, anaemia, arterial route.
- **Prevention: IV fluid before contrast** — only effective measure; †for eGFR <30 or AKI; lowest dose. Post-contrast dialysis never needed.
- **Gadolinium**: groups I and III → **NSF** — avoid in AKI, eGFR <**30**, dialysis. Group II (gadobenate, gadoteridol, gadobutrol, gadoterate) allowed. **Group III (gadoxetate) treated as group I.**

Acute interstitial nephritis

- **Drugs cause most**: β-lactams, quinolones, sulfonamides, vancomycin, NSAIDs, PPIs, allopurinol, **checkpoint inhibitors** (commonest biopsy lesion in their AKI).
- Days–weeks after drug; think of it in unexplained hospital AKI with **sterile pyuria**. Urine eosinophils insensitive, non-specific MCQ.
- **Stop the drug**; early steroids (observational); ≈half recover fully.

Glomerulonephritis

- In the microhaematuria differential: SCr rise, proteinuria, red cell casts, hypertension, purpura → **urgent nephrology + biopsy**, not cystoscopy first (RPGN). Nephrotic disease insidious.

Complications of AKI

- **Overload** (urgent only with respiratory distress): salt/fluid restriction, high-dose IV loop, redose by half-life; add metolazone or acetazolamide (**sequential blockade**). Oral furosemide = 2 × IV dose; torsemide longest (8–12 h).
- Uraemic bleeding → **DDAVP 0.3 µg/kg** IV/IM 30–60 min pre-procedure; tachyphylaxis after 3 doses; watch hyponatraemia.

Renal replacement therapy

- AEIOU = failure of medical management, not a biochemical threshold. **Diffusion** clears solute, **convection** fluid. **Early dialysis no benefit; continuous not superior.**
- Non-recovery ≥ **90 days** = CKD/ESKD.

4 · Drug dosing in reduced kidney function

- Dose on steady-state eGFR; in AKI eGFR overestimates — reassess daily. †**Cockcroft–Gault** = (140 − age) × weight (kg) / (72 × SCr mg/dL), × 0.85 if female.
- Avoid NSAIDs; †withhold RAAS blockers in dehydration/perioperative hypotension ("sick-day" rule).
- †Nitrofurantoin: avoid if eGFR <30 (short courses at 30–44, UK). †Metformin: stop at eGFR <30; withhold before iodinated contrast in AKI or eGFR <30.

5 · Chronic kidney disease

- **KDIGO**: abnormal structure or function ≥ **3 months**; structure alone qualifies (atrophic kidney, normal GFR). Classify by **cause, GFR, albuminuria**.

GFR (mL/min/1.73 m²)	A1 ACR <30 mg/g (<3 mg/mmol)	A2 30–300	A3 >300 (>30 mg/mmol)
G1 ≥90	Low*	Moderate	High
G2 60–89	Low*	Moderate	High
G3a 45–59	Moderate	High	Very high
G3b 30–44	High	Very high	Very high
G4 15–29	Very high	Very high	Very high
G5 <15 (kidney failure)	Very high	Very high	Very high

*G1–G2 are CKD only with albuminuria or structural damage. Albuminuria can shift two risk bands without GFR change: eGFR 43, ACR 1100 mg/g = **G3bA3, very high**.

- Causes: diabetes, hypertension, GN, **stones, chronic obstruction**, genetic. **Each perioperative AKI is a modifiable progression factor**, as are albuminuria, smoking, acidosis.
- P2 Metabolic abnormalities in CKD — *complications table: anaemia (Hb **10–11.5 g/dL**, >13 harmful), secondary/tertiary HPT, acidosis (bicarbonate ≥**22**), hyperkalaemia (binders, keep RAAS blockade), hypertension (chlorthalidone), ED*
- Tertiary HPT: calcimimetic → subtotal parathyroidectomy. ED: PDE5 inhibitors, transplant.

Slowing progression

- HbA1c <7.5%; BP <**130/80**; agent matters only in proteinuric disease → RAAS blockade.
- **SGLT2 inhibitors**: reno-/cardioprotection, mortality benefit, ± diabetes. **Finerenone** in CKD + type 2 diabetes.

6 · Haematuria and proteinuria — when to involve a nephrologist

- †Microhaematuria ≥**3** RBC/HPF on microscopy; dipstick needs confirmation.
- **Glomerular clues**: dysmorphic RBCs, red cell casts, raised ACR/PCR, rising SCr, hypertension, purpura/systemic disease. †Glomerular disease does not exclude urothelial cancer.

> **ALGORITHM**
> Haematuria ± proteinuria → microscopy + ACR + SCr/eGFR → casts, dysmorphic cells, AKI or systemic features → **urgent nephrology**; isolated non-glomerular haematuria → urological evaluation; isolated albuminuria → stage CKD, refer per criteria.

- †KDIGO refer: AKI/abrupt GFR fall; GFR <30; ACR ≥300 mg/g (≥30 mg/mmol); fall >5 mL/min/1.73 m²/year; red cell casts or haematuria >20 RBC/HPF; BP refractory to ≥4 drugs; persistent K abnormality; recurrent stones; hereditary disease.

7 · Electrolyte and acid–base disorders

Water and sodium

- **Sodium disorders are water disorders**; volume is a separate clinical call.
- ADH (V2 → **AQP2**): **volume depletion overrides hypo-osmolality**.
- Of 1 L: D5 → **83 mL** intravascular; 0.9% saline 250 mL; 5% albumin 1000 mL.

Hyponatraemia

- Confirm hypo-osmolality; hyperglycaemia → correct Na. †TUR syndrome: hypotonic glycine/sorbitol absorption in monopolar resection.

Volume	Urine	Causes
Hypo	Na >20	Diuretics, mineralocorticoid deficiency, cerebral salt wasting
Hypo	Na <20	Vomiting, diarrhoea
Eu	Osm > serum	SIADH, cortisol deficiency, hypothyroidism, pain, drugs
Eu	Osm low	Polydipsia, low solute ("tea and toast", beer potomania)
Hyper	Na >20	Kidney failure
Hyper	Na <20	Heart failure, cirrhosis, nephrotic

- SIADH (exclusion) → water restriction → vaptans, salt + furosemide, urea.

- **Rate**: chronic → ≤**8–12 mEq/L per 24 h**, else **osmotic demyelination**. Highest risk: resuscitated hypovolaemia → water diuresis; overcorrection → DDAVP + free water. Severe symptoms → **3% saline to raise 4–6 mEq/L**, then slow.

Hypernatraemia

- Nephrogenic DI: lithium, hypercalcaemia, hypokalaemia, **post-obstruction**; nil-by-mouth DI patient at risk.
- Free water; chronic → fall ≤**0.5 mEq/L/h** (else cerebral oedema).

Potassium

- P1 Management of acute hyperkalaemia — *98% intracellular; causes incl. obstructive uropathy, jejunal conduit, pseudohyperkalaemia; ECG sequence; >**6.0** urgent; binders (SZC ≈1 h, patiromer ≈7 h)*

> **ALGORITHM**
> **Stabilise** (any ECG change): calcium gluconate 10% 10 mL IV → **shift**: insulin 10 U + 50 mL 50% dextrose; nebulised salbutamol 20 mg → **remove**: haemodialysis if oliguric/ESKD; loop diuretic if non-oliguric; bicarbonate if acidotic; binder → stop culprit drugs.

- **SPS**: only emergency-approved binder, yet slow and causes **colonic necrosis** (postoperative, ileus) → not first-line.
- **Hypokalaemia**: **Vomiting** → alkalosis → bicarbonaturia → distal Na delivery → kaliuresis (not gastric loss). KCl, **oral preferred**, 20–40 mEq repeated.

Calcium

- Saline/loop → calciuria; **thiazide ↓ urine calcium** (DCT, Na-uncoupled). Corrected Ca: +0.8 mg/dL per 1 g/dL albumin below 4.
- **Hypercalcaemia, stone former**: PTH non-suppressed = primary HPT; suppressed → PTHrP (renal, bladder, squamous cancer), 1,25-vitamin D (sarcoid, lymphoma). Low urine Ca → FHH.
- Treat if symptomatic or ≥**14 mg/dL**: **isotonic saline** 200 mL/h to urine 100–150 mL/h; loop only if overloaded; calcitonin (rapid, tachyphylaxis 48 h); IV bisphosphonate; denosumab.

Acid-base

- AG = Na − (Cl + HCO_3), raised >12; +2.5 per 1 g/dL albumin below 4. AG rise < HCO_3 fall → coexisting non-AG acidosis.
- **Urine AG** = (Na + K) − Cl: negative = ammonium = gut loss; positive = renal (RTA).
- P2 Renal tubular acidosis — *type 1/2/4 table: defect, K, urine pH, stones; obstruction → type 4; UAG*
- Proximal: hypokalaemia worse with HCO_3; urine pH high on carbonic anhydrase inhibitors; **topiramate, zonisamide** → CaP stones.
- Hyperkalaemic distal RTA: **obstruction**, lupus, sickle cell, amiloride, triamterene, trimethoprim. 1 mmol citrate = 1 mmol HCO_3.
- MCQ Laxative abuse → negative urine AG, urine pH >5.8 → **ammonium acid urate stones**. †Ileal/colonic conduits, neobladders → hyperchloraemic non-AG acidosis (NH_4Cl absorbed).

8 · Urinary tract obstruction and post-obstructive diuresis

Term	Meaning
Hydronephrosis	Pelvicalyceal dilatation ± obstruction
Obstructive uropathy (OU)	Obstruction at any level
Obstructive nephropathy	Irreversible parenchymal loss

- OU is **curable** — exclude it (†ultrasound) in unexplained AKI/CKD; unilateral may not raise SCr → *Paediatric Urology book, Ch 51*.
- P1 Post-obstructive diuresis — *after bilateral obstruction relief; up to **300 mL/h**, ≈2 days; 3 mechanisms (AQP2/Na-K-ATPase loss, volume expansion/ANP, urea); gradual decompression no role*
- Transporter suppression may last weeks → nephrogenic DI (urine 300–400 mOsm/kg, hypernatraemia).

ALGORITHM

Relieve obstruction → free oral fluids, no routine IV → persistently high output, hypovolaemia or obtunded/nil by mouth → panel every **4–6 h** (Na, K, phosphate, Ca, Mg) → replace ≈**75%** of output with **0.45% saline** + electrolytes.

KEY POINT

Why not replace all output? The diuresis is mostly appropriate excretion of retained salt, water and urea; †full replacement keeps the patient expanded and sustains it.

Recovery after obstruction

Canine study	Model	Recovery
Vaughan and Gillenwater	Complete UUO	7 days → 100%; 14 days → 70%; 4 weeks → 30%; **6 weeks → none**

- Injury **vascular** before tubular; duration, not dilatation, governs it.
- Small stone, low-grade obstruction: observe **2–4 weeks** safely; beyond 4 weeks unpredictable (recovery up to 5 months).
- Recovery not judgeable acutely → **low threshold for stent/nephrostomy**. †Infection + obstruction → urgent drainage.

9 · Landmark trials

Trial	Design	Result	Take-home
†PRESERVE	Bicarbonate vs saline, NAC vs placebo	No difference	Saline; no NAC
†STARRT-AKI	Accelerated vs standard RRT	No survival benefit; more dialysis dependence	Dialyse for AEIOU, not early
†DAPA-CKD / EMPA-KIDNEY	SGLT2i, CKD ± diabetes	Less progression	Standard in CKD
†FIDELIO-DKD	Finerenone, CKD + T2DM	Fewer kidney and CV events	Add to RAAS blockade
†CHOIR / TREAT	High vs lower Hb target	Harm with high target	Avoid Hb >13 g/dL

91 Renal Transplantation

1 · End-stage disease and why transplant

- KDIGO **G5 (eGFR < 15 mL/min/1.73 m²)** = ESRD → *Ch 90*.
- Dialysis years add cardiovascular and mineral–bone damage → **transplant is optimal therapy**; **pre-emptive** (before dialysis) best — commoner with living donors and children.
- May triple life expectancy; cheaper after year one. AV access still created ahead of need. Death risk **higher for first 3–6 months**, then much lower (Wolfe).
- MCQ First successful kidney transplant: identical twins, 1954.

2 · Recipient evaluation

> **KEY POINT**
> Two questions: will the patient survive the operation, and will immunosuppression unmask something dormant (infection, cancer)?

- Evaluate before dialysis; deceased-donor listing at eGFR ≤ 20.

Recurrence risk	Diseases	Implication
Very low	ADPKD, dysplasia, Alport, **genetic** FSGS	None
Higher	**Primary FSGS**, MPGN, membranous	Counsel; monitor
Histological often, clinical rarely	IgA nephropathy	—
Metabolic	Primary hyperoxaluria	Combined liver–kidney

Native-gene disease cannot recur in a donor-gene graft; circulating/immune disease can.

- **Cardiac**: CAD = main cause of death. High risk (diabetes, smoking, prior CAD, long dialysis) → echo, stress test, selective angiography.
- **Vascular**: heavy iliac calcification may preclude anastomosis → Doppler, non-contrast CT. Non-healing foot ulcer → defer. ADPKD + family history of intracranial bleed → aneurysm screen.
- **Obesity** (relative contraindication): wound complications, DGF, rejection. Frailty predicts mortality. Non-adherence, substance abuse must be controlled.
- **Infection**: active TB contraindicates until treated; HBV, HCV, HIV; CMV/HSV/EBV serology guides prophylaxis. **Vaccinate before** — no live vaccines after. Antivirals allow HCV-viraemic kidneys into negative recipients.

Malignancy	Transplant
Prostate Gleason ≤6; superficial non-melanoma skin cancer; incidental renal tumour <1 cm	Without prior treatment
Basal/squamous skin, melanoma in situ, RCC <3 cm, small low-grade thyroid, superficial bladder cancer	No wait after cure
All other	After treatment, individualised wait

- Long-term femoral dialysis access → exclude iliac venous obstruction; ipsilateral leg fistula → graft venous hypertension.

- P2 Pre transplant nephrectomy — *indications: ADPKD space/infection/bleeding/pain, chronic infection, VUR with UTI, heavy proteinuria*

3 · Immunological compatibility

- P3 Discuss the role of tissue typing and cross match in live and cadaveric — *ABO rules (A2 donors), HLA class I/II, sensitisation, crossmatch, Luminex virtual crossmatch, paired exchange*

4 · Deceased donation

- P2 P1 Deceased (cadaver) kidney donor: selection, identification, maintenance, distribution and retrieval; Kidney retrieval techniques for organ donation — *DBD vs DCD, contraindications, KDPI/EPTS, en bloc <**15 kg**, retrieval steps, UW/HTK, machine perfusion ↓DGF*
- En bloc into adult with BMI <30, low immunological risk; thrombosis 10–15%, long-term survival = living donation.
- Cold storage at 4 °C slows (not stops) metabolism; machine perfusion acts on ischaemia–reperfusion injury, no long-term graft benefit.

	DBD	DCD
Warm ischaemia	Minimal	Present → rapid aortic cannulation, cooling
Preservation	Static cold storage	Usually machine perfusion (↓DGF)

ALGORITHM

Procurement: midline + sternotomy → bowel off retroperitoneum → control aorta, IVC → find aberrant/accessory arteries **before cannulation** → perfuse; ice via opened Gerota → **periureteric tissue preserved** → en bloc → back-table separation, cold storage

5 · Living donation

- P2 Pre-operative evaluation of the living (related and unrelated) renal donor — *donor safety, exclusions (bilateral stones), GFR by 2 methods, CT angiography, split function; function 60–80%, ESKD 90 vs 14/10,000*
- Donor mortality **3/10,000**; more gestational hypertension/pre-eclampsia; independent donor advocate mandatory.
- **Laparoscopic donor nephrectomy** (Ratner 1995) standard. Accessory arteries may contraindicate it → *Ch 88*.

6 · The operation

- ABO check before induction. Back table: ligate vein tributaries and hilar lymphatics (lymphocele); **never dissect the lower pole–ureter triangle**; extend short right vein with donor IVC.
- **Multiple arteries** (18–43%): more DGF, similar survival; Carrel patch, conjoining, reimplantation or separate anastomoses. **Always keep the lower pole artery** (ureter supply); upper pole branches ≤2 mm may be ligated.
- Retroperitoneal iliac fossa (right traditional). **Contralateral placement** puts pelvis and ureter anterior.

	Gibson	Hockey stick
Course	Oblique, 2 cm above pubis to near ASIS	Midline suprapubic, up rectus edge
Access	Iliac vessels	Better to common iliac, IVC

	Gibson	Hockey stick
Hernia	Lower	Higher; rectus denervation

- Nerves at risk: lateral femoral cutaneous (near ASIS), femoral (retractor), genitofemoral.

> **ALGORITHM**
> **Vein first**: end-to-side external iliac vein, 5-0/6-0 polypropylene → artery end-to-side external iliac (punch, avoid calcified clamp sites) → reperfuse → position without kink or stretch

- **Hume test**: finger compression of renal vein → pulsatile swelling, relieved on release.
- Flaccid (inflow) or tense engorged (outflow) graft → check, intraoperative ultrasound.
- Avoid internal iliac if contralateral occluded (gluteal claudication, ED).

> **ALGORITHM**
> **Lich–Gregoir** extravesical: fill bladder → myotomy → open mucosa distally → spatulate → mucosa-to-mucosa 5-0/6-0 absorbable ± stent → loose detrusor closure as short tunnel

- Document stent removal — forgotten stents encrust. Urological complications → *Ch 92*.

7 · Surgical complications

- P1 Vascular complications after renal transplantation, including transplant renal artery stenosis — *arterial/venous thrombosis (reversed diastolic flow), TRAS (parvus-tardus, angioplasty first), thrombosis risks, torsion, pseudoaneurysm, biopsy AV fistula*
- Bleeding: brisk drain, falling haematocrit → transfuse, re-explore.
- P1 Lymphocele and postoperative lymphatic leak (chylous ascites) — *risks incl. mTOR inhibitors; lymph creatinine = serum; treat if symptomatic; laparoscopic fenestration best, aspiration worst*

Transplant nephrectomy

- For **graft intolerance syndrome** (pain, swelling, haematuria, fever — steroids first), infection, tumour; not asymptomatic failed grafts.
- **Early (<6–8 weeks) extracapsular**; late **subcapsular** — except tumour.

8 · Rejection

- Induction → tacrolimus + mycophenolate ± steroids.
- P1 Renal allograft rejection. — *hyperacute/acute/subclinical/chronic; acute <10%, chronic AMR limits survival; biopsy (Banff TCMR/AMR); risks; de novo DSA, dd-cfDNA NPV >90%; TCMR steroids → ATG; AMR plasmapheresis, IVIG, rituximab, eculizumab, anti-IL-6*
- Hyperacute: †irreversible → graft nephrectomy. De novo DSA linked to class II mismatch and under-immunosuppression.

9 · Landmark trials

Trial	Design	Result	Take-home
Wolfe 1999	Registry: transplant vs wait-listed dialysis	Risk 2.84 early; equal at 106 days; long-term 0.32	Early peril, durable survival gain

★ Putting it together: kidney function and transplant decisions

A · Rising creatinine after urological surgery

ALGORITHM

1. Postrenal first: catheter, bladder/renal scan. After cystectomy/bladder repair, rise + ascites = possible urine leak; drain creatinine ≫ serum (→ *Ch 90*).

2. Prerenal vs intrinsic: FeNa <1%, bland urine → volume depletion; >1% + granular casts → ATN. Stop nephrotoxins, RAAS blockers, metformin.

3. Dialysis only for refractory "AEIOU": acidosis, electrolytes (K), intoxication, overload, uraemia.

B · Post-obstructive diuresis

KEY POINT

After bilateral (or solitary-kidney) relief monitor output, electrolytes. Mostly physiological; †pathological if it persists after euvolaemia. †Don't replace every millilitre. If unable to drink, replace ≈75% with 0.45% saline (→ *Ch 90*).

C · Transplant ureteric complications

Problem	Clue	Management
Urine leak (early)	High drain output; fluid creatinine ≫ serum	Foley + nephrostomy ± stent; explore if large, proximal, persistent
Ureteric stricture	Falling function ± hydronephrosis (may be mild)	Nephrostomy → balloon or endoureterotomy; open repair (reimplant, ureteroureterostomy, Boari) most durable
Lymphocele	Fluid creatinine ≈ serum; may obstruct ureter/vessels	Only if symptomatic (<15%); laparoscopic peritoneal window best; aspiration recurs
Stent-related UTI	Colonised stent	Remove at 14–21 days (earlier → leak; later → infection)

Prevent: stent; spare lower pole–ureter triangle; reimplant lower polar arteries (→ *Ch 91, 92/61*).

D · Rejection at a glance

Type	Cause	Point
Hyperacute	Preformed DSA	Crossmatch prevents; †irreversible → graft nephrectomy
Acute	TCMR or AMR (Banff)	Biopsy. TCMR: IV steroids → ATG. AMR: steroids, IVIG, plasmapheresis, rituximab
Chronic	Mainly antibody-mediated	Main late loss; subclinical

E · Child going for transplant

ALGORITHM

Mostly CAKUT, chiefly valves. Make the bladder safe: age-appropriate capacity at detrusor pressure <40 cm H_2O (antimuscarinics, CIC, augmentation if needed). Remove refluxing ureters/kidneys causing infection, polyuria, hypertension. A poor bladder destroys the graft (→ *Ch 92/61*).